

Merger Arbitrage and Synergy Valuation:

The Sun Pharma – Organon Deal

A standalone valuation, event study, and merger-arbitrage risk assessment

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Abstract

On 26 April 2026, Sun Pharmaceutical Industries agreed to acquire NYSE-listed Organon & Co for \$14.00 per share in cash, a 102.9% premium to the last unaffected close and the largest deal ever undertaken by an Indian pharmaceutical company. This report values Organon on a standalone basis, tests the market's reaction around the deal's key dates using an event-study framework, and assesses the risk-adjusted return available to a merger-arbitrage position bought at the current spread. The standalone discounted cash flow value, calibrated to and cross-checked against the market's own pre-deal pricing, implies the \$14.00 offer requires Organon's revenue to reverse from a two-year decline to roughly 3.9% annual growth - a real but plausible assumption, equivalent to an implied synergy premium of approximately \$1.9 billion. The acquirer's own stock reacted positively and significantly to the announcement, arguing against a simple overpayment narrative. On the arbitrage question, a forward-looking Monte Carlo simulation indicates the position's expected annualised return, at a conservatively assessed completion probability, falls below the risk-free rate - a conditional, not categorical, recommendation follows. Every figure in this report is traceable to a Python analysis pipeline built for this purpose, cross-checked with SEC filings and market data, with material methodological choices - the discount-rate selection, the treatment of non-overlapping trading hours between NSE and NYSE, and the volatility regime used for hedge pricing - stated and justified explicitly rather than assumed.

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1. Introduction

Sun Pharmaceutical Industries Limited, India's largest pharmaceutical company by market capitalisation, agreed on 26 April 2026 to acquire Organon & Co, a US-listed women's health and biosimilars company spun off from Merck in 2021, for \$14.00 per share in an all-cash transaction. The deal values Organon's enterprise at \$11.75 billion and is the largest acquisition ever undertaken by an Indian biopharmaceutical company. Sun Pharma's own 8-K filing frames the transaction as generating "a stronger cash generating company" with post-transaction net debt to EBITDA of 2.3x, and cites "scope for synergies including significant revenue upside opportunities" without disclosing a dollar figure.

This report addresses three linked questions. First, is \$14.00 a fair price for Organon on a standalone basis, and what does the premium imply about the growth or synergies the market is being asked to underwrite? Second, how did the market actually behave around the deal's key dates - the period of speculation that preceded the formal announcement, the announcement itself, and the shareholder vote - and what does that behaviour reveal about how the deal was priced? Third, given the deal remains open as of the snapshot date, does the arbitrage spread available today compensate for the risk that the transaction does not close?

Organon is a useful subject for this kind of analysis precisely because it is a difficult one: an 11.5x levered, margin-compressing, patent-cliff-exposed business carrying a capital structure closer to a leveraged issuer than a typical pharmaceutical company. That combination makes several standard shortcuts - using a company's historical average cost of debt, applying a single point-estimate discount rate, or pricing options off a company's own listed quotes - actively misleading here, and each of those failure modes is confronted directly in Section 3.

2. Literature Review

Three strands of the M&A literature motivate the analytical choices in this report.

Roll (1986) proposes the hubris hypothesis: acquiring managers infected by overconfidence pay a positive-valuation-error premium for targets, and the resulting overpayment should show up as a negative or muted market reaction in the acquirer's own stock around the announcement. Andrade, Mitchell and Stafford (2001), surveying merger activity through the 1990s, find that the combined announcement-period reaction across acquirer and target is typically positive, but that the acquirer's own reaction on average is close to zero or mildly negative - the hubris prediction holds as a base rate even where it does not hold in every individual deal. This report tests that base rate directly against Sun Pharma's own announcement-window abnormal return (Section 4.2) rather than assuming it.

Jensen (1986) offers a complementary agency-cost explanation for value-destructive acquisitions: managers of firms generating free cash flow in excess of positive-NPV internal investment opportunities have an incentive to deploy that cash on acquisitions, including value-destructive ones, rather than return it to shareholders. This framing is relevant to Organon

specifically in reverse - as the target, not the acquirer - since Organon's own capital structure (heavy debt, a recent divestiture, compressing margins) is more consistent with a business under balance-sheet pressure than with one flush with discretionary cash, which shapes how the standalone valuation in Section 4.1 treats its cash flow generation.

Mitchell and Pulvino (2001), analysing 4,750 mergers between 1963 and 1998, characterise merger-arbitrage returns as resembling the payoff of selling uncovered index put options: positively correlated with the market in severely depreciating conditions, uncorrelated in flat or rising markets, and - after accounting for that nonlinearity and transaction costs - earning roughly four percent per year in excess returns. That framing directly informs the approach in Section 4.3: rather than reporting a single point-estimate expected return, the arbitrage position's risk is characterised through its full distribution, including its tail, which is where an options-like payoff structure matters most.

Cost-of-capital and credit-risk inputs draw on Damodaran's regularly updated, publicly available datasets rather than static textbook assumptions: the implied (forward-looking) US equity risk premium, and the mapping from a company's interest coverage ratio to a synthetic credit rating and default spread for firms without a traded public bond rating. Both are used and cited specifically in Section 4.1.

3. Data and Methodology

3.1 Data sources

Daily price and volume history, and annual and quarterly financial statements, for Organon (OGN), Sun Pharma (SUNPHARMA.NS), the S&P 500, the Nifty 50, the VIX, the USD/INR exchange rate, the 10-year US Treasury yield, and a peer set of four US pharmaceutical companies (Merck, Pfizer, Viatris, Teva) and three Indian pharmaceutical companies (Cipla, Dr Reddy's, Lupin) were pulled from Yahoo Finance and persisted as a single dated snapshot, so that every figure in this report traces to one fixed pull rather than drifting across re-queries. Deal terms and the confirmation that no dollar synergy figure was disclosed were taken directly from Organon's Form 8-K, Exhibit 99.1 (filed 27 April 2026, SEC EDGAR accession 0001193125-26-178718) - read in full, not inferred from secondary reporting. The equity risk premium and the interest-coverage-to-credit-rating mapping used in the discount-rate work were taken from Damodaran's publicly maintained datasets (Section 2).

3.2 Standalone valuation

Free cash flow to the firm is built from each year's EBIT, a normalised (not raw effective) tax rate, depreciation and amortisation, capital expenditure, and the change in net working capital. The raw GAAP effective tax rate was tested first and rejected: it swings from +56% to -52% year to year on valuation-allowance and one-off items unrelated to the operating business, which would make free cash flow non-comparable across years for reasons that have nothing to do with performance.

Cost of equity follows CAPM, with beta estimated over a 250-trading-day window ending the day before 16 January 2026 - deliberately excluding the run-up that followed, since a window anchored to the 26 April announcement would sit on top of that run-up and contaminate both alpha and beta with the event itself. Cost of debt is estimated two ways: an embedded rate (interest expense over average debt, reflecting Organon's 2021 spinoff-era, healthier-credit terms) and a marginal rate (risk-free rate plus the default spread implied by Organon's current 1.84x interest coverage under Damodaran's synthetic-rating framework, which maps to a B3/B-rating). The two bound a wide range. Rather than choose between them, a third rate is solved numerically: the discount rate at which the standalone DCF, on a flat-revenue base case, reproduces the actual \$6.90 unaffected market price. That market-implied rate falls inside the embedded-to-marginal range, which is treated as validating the framework, and it is the rate used throughout the valuation work that follows.

The explicit five-year forecast holds revenue flat and every other driver at its FY2025 ratio to revenue, terminal growth at 1.5%. A reverse-DCF then solves for the constant revenue growth rate that makes the same model reproduce the \$14.00 offer price, which is read as the growth assumption - and, by difference against the standalone enterprise value, the implied synergy premium - the offer requires.

3.3 Event study

Abnormal returns follow the standard market-model approach: alpha and beta are estimated on the same pre-leak, 250-day window used for the valuation beta, and cumulative abnormal returns are computed over $(-1,+1)$, $(-5,+5)$ and $(-10,+10)$ day windows around three dates - the 16 January 2026 reference date, the 26 April 2026 announcement, and the 23 July 2026 shareholder approval - separately for Organon (the target) and Sun Pharma (the acquirer), with significance assessed via the estimation-window residual standard deviation. A parallel test checks for abnormal trading volume around the reference date, and a peer-spillover panel tests whether the four US pharmaceutical peers showed any comparable reaction.

Every single-ticker return series in this analysis is built directly from that ticker's own trading calendar, not from a panel indexed on the union of every ticker's dates. That distinction matters: a panel built across NSE- and NYSE-listed instruments necessarily includes dates on which only one of the two markets traded, and computing a return by shifting one row on that unioned index - rather than one actual trading day - silently drops or corrupts real observations whenever such a date falls between two sessions for a given ticker.

3.4 Volatility and cross-market causality

A GARCH(1,1) model is fit separately on the pre- and post-announcement samples, since a single model spanning the 26 April break would average together two different volatility regimes. Granger causality between Organon and Sun Pharma is tested in both directions, but the two directions are not symmetric by construction: NSE trades roughly 03:45-10:00 UTC and NYSE roughly 13:30-20:00 UTC, so Sun Pharma's same-calendar-date close is genuinely prior information relative to Organon's same-date close, while the reverse is not true until the following trading day. The test of whether Organon's past returns predict Sun Pharma's uses standard lagged Granger causality without adjustment; the reverse test is reported both in the conservative lagged form and as a same-day OLS test that is legitimate given the verified non-overlap in trading hours, though it is not classical Granger causality in the textbook sense since it uses a contemporaneous rather than lagged regressor.

3.5 Merger-arbitrage sizing and hedging

The spread-implied completion probability follows the standard practitioner identity, $p = (\text{spot} - \text{break}) / (\text{offer} - \text{break})$, evaluated across three break-price scenarios (the unaffected price, and 10% and 20% below it) since the break price is a proxy, not a measurement. A forward-looking Monte Carlo simulation (100,000 paths, seeded for reproducibility) draws completion probability, break price, and time to close from independent distributions and reports the resulting holding-period return distribution in full, including its Value-at-Risk and Conditional Value-at-Risk tails.

Organon's own listed options were tested directly for usability before being ruled out: every expiry checked showed a zero bid and zero ask at the money, and the same pattern held across every large US pharmaceutical peer checked, not Organon alone - this is a data-feed limitation,

not company-specific information. Black-Scholes prices and Greeks are therefore built from first principles, using realised rather than implied volatility, and specifically the pre-leak volatility regime rather than the post-announcement (deal-pinned) regime - using the latter would price deal-break insurance using the very volatility suppression that only holds if the deal in fact closes, which is circular.

4. Findings

4.1 Standalone valuation

Organon's capital structure shapes every part of this valuation. Against FY2025 EBIT of \$929 million, net debt stood at \$8.07 billion - a net debt / EBITDA ratio of 6.26x and an interest coverage ratio of 1.84x, both far outside the range of the US and Indian pharmaceutical peers examined (Merck's debt/equity is 0.94x, Pfizer's 0.74x; the least levered peer, Cipla, sits at 0.02x). EBIT margin compressed from 25.0% in FY2022 to 14.95% in FY2025, and revenue itself was down 2.9% in FY2025 alone.

Discount rate estimate	Value	Basis
Embedded cost of debt	5.29%	Interest expense / average debt (2021 spinoff-era terms)
Market-implied (used)	6.61%	Solved to reproduce the \$6.90 unaffected price
Marginal / synthetic-rating	7.90%	Risk-free + Damodaran B3/B- default spread

The market-implied rate falling cleanly inside the embedded-to-marginal range is treated as validating the discount-rate framework, not as circular reasoning: the embedded and marginal estimates are each independently constructed from Organon's own financials and credit profile, without reference to the traded share price, and neither was built to hit any particular target.

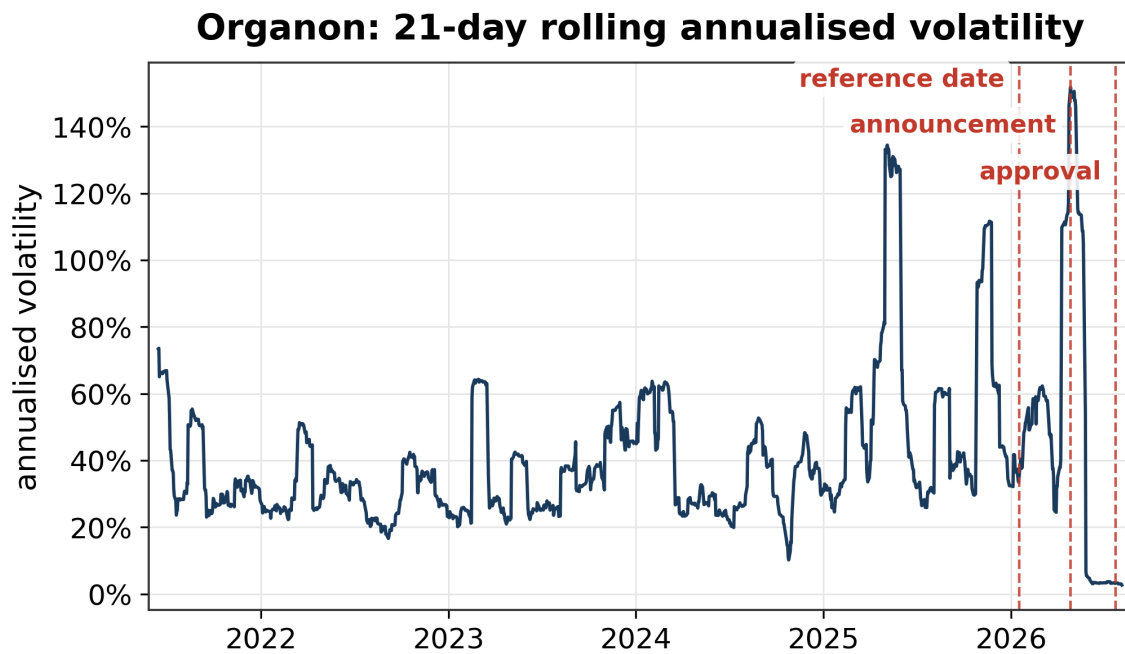
Metric	Value
Standalone equity value / share (flat-revenue base case)	\$6.90
Offer price	\$14.00
Revenue growth required to justify the offer (constant, 5yr)	3.87%
Actual FY2023-FY2025 revenue CAGR	-0.38%
Implied synergy / growth premium	\$1.86 billion (18.9% of standalone EV)
Terminal value as % of enterprise value	77.7%

A required growth reversal from -0.38% to +3.87% is a real ask, but not an implausible one - it does not require Organon to become a growth business, only a stable one, and the deal narrative (a recently closed divestiture, folding into a larger commercial platform) is consistent with that kind of stabilisation. The terminal value's 77.7% share of enterprise value is flagged

rather than smoothed over: this is a mature, low-growth business, and most of its value legitimately sits in the terminal period rather than the explicit forecast.

4.2 Market behaviour around the deal

Realised volatility moved through four distinct regimes: a 44.8% multi-year baseline, 58.0% in the second half of 2025 as speculation began building, 94.2% in the window between the 16 January reference date and the announcement, and a collapse to 4.5% once the deal was signed - a roughly 95% reduction, consistent with a stock now priced almost entirely off completion risk rather than business fundamentals.



Source: Yahoo Finance / SEC EDGAR

Figure 1. Organon 21-day rolling annualised volatility, with deal-timeline markers.

69.6% of the total price move from the unaffected price to the post-announcement price (on a simple-return basis; 75.8% on a log-return basis) occurred before the formal 26 April announcement. That finding is real, but it is weaker evidence of a discrete leakage event than it first appears: the narrow-window cumulative abnormal return around the 16 January reference date is not statistically significant at conventional levels (CAR = +14.3% over a ± 5 -day window, $t=1.09$, $p=0.30$), and there is no abnormal trading volume around that date either (event-window volume was 1.04x the 60-day baseline, $z=0.05$). The balance of evidence supports a large, gradual re-rating accruing over roughly three months of building speculation, not a sharp, detectable jump concentrated on a single day.

Around the announcement itself, Organon's cumulative abnormal return was large and statistically significant (CAR = +33.0% over ± 5 days, $t=2.52$, $p=0.03$) - unsurprising for the target. More informative is Sun Pharma's own reaction: also positive and statistically significant (CAR =

+9.6% over ± 5 days, $t=2.64$, $p=0.03$). Under the hubris-hypothesis base rate (Section 2), an acquirer paying a large premium for a declining, highly levered target would typically show a muted or negative reaction; Sun Pharma's positive, significant reaction argues against a simple overpayment narrative, and corroborates the valuation finding in Section 4.1 that the growth assumption embedded in the offer is a plausible ask rather than an extreme one. None of the four US pharmaceutical peers tested showed a statistically significant reaction around the announcement, indicating the deal's information content was specific to these two companies rather than a read-through for the sector.

Cross-market causality is asymmetric in a way consistent with Organon being the deal-specific driver: Organon's own past returns Granger-cause Sun Pharma's returns at every lag tested (lag-1 $p=0.0001$), while Sun Pharma's returns do not predict Organon's, whether tested with a conservative one-day lag ($p=0.57$) or, exploiting the two markets' non-overlapping trading hours, on a same-day basis ($p=0.98$).

4.3 Merger-arbitrage risk assessment

At the snapshot-date spot price of \$13.57 against the \$14.00 offer, the gross spread is 3.17%, implying a market-assessed completion probability of 93.9% on the unaffected-price break-price basis (rising to 94.9% under the more conservative -20% break-price scenario). That is a high implied probability, consistent with a deal that has already cleared shareholder approval.

Monte Carlo outcome distribution, bought at \$13.57 (N=100,000)

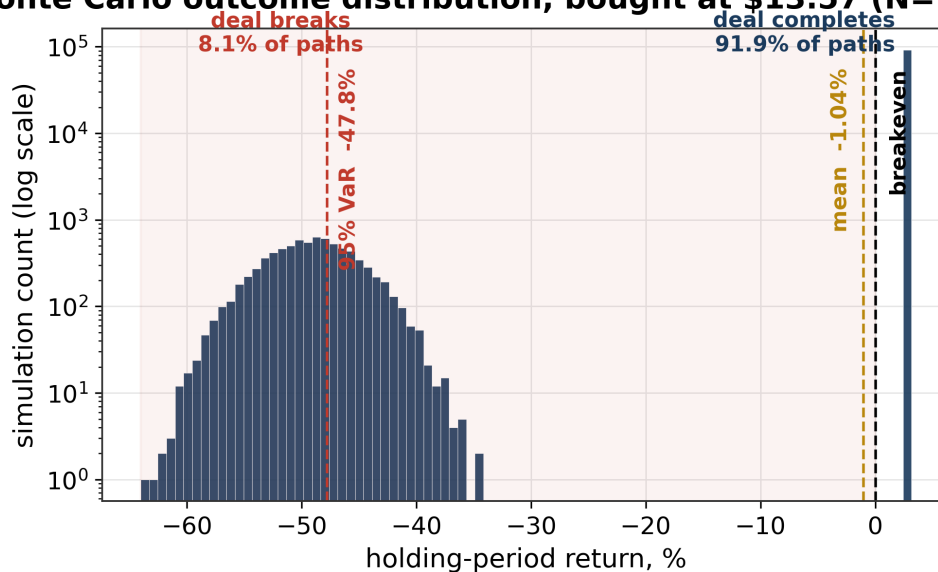


Figure 2. Monte Carlo holding-period return distribution, 100,000 simulated paths, position bought at today's spot price.

Monte Carlo statistic (100,000 paths)	Value
Mean holding-period return	-1.04%

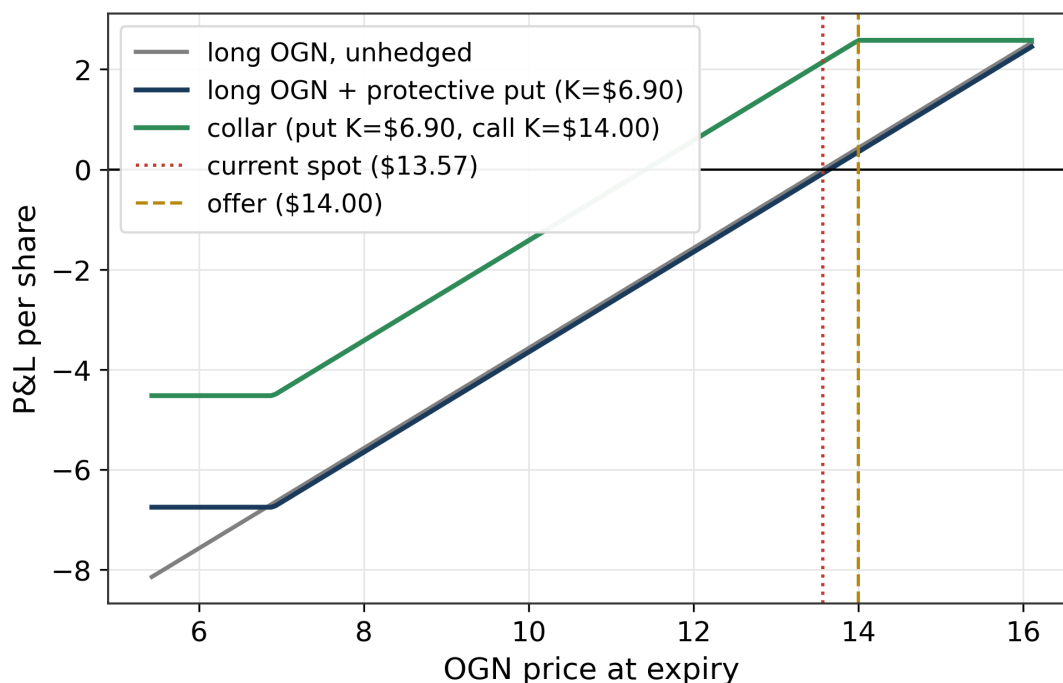
Mean annualised return	0.07% (below the 4.62% risk-free rate)
Probability of a loss	8.1%
95% VaR	-47.8%
95% CVaR	-51.5%
Max-Sharpe two-asset allocation (arb position vs. S&P 500)	0% arb / 100% index

This is the central, and least intuitive, finding of the risk assessment. A backward-looking Sharpe ratio calculated over the period since the announcement is high (2.10) - but that figure is dominated by the one-time +16.9% announcement-day jump, which has already happened and cannot be earned again by a position opened today. Buying at today's price is a different bet: the simulation's ~8% assessed break probability, applied to a loss of roughly half the position if the deal fails, outweighs the 3.17% gross spread even though completion remains the more likely single outcome. On these assumptions, the position bought today does not clearly compensate for the risk being taken.

4.4 Hedging deal-break risk

Before pricing any hedge, Organon's own listed options were checked directly rather than assumed usable: every expiry showed a zero bid and zero ask at the money, with zero open interest at the nearest expiry, and the same pattern held for Merck, Pfizer, Viatris and Teva - a data-feed limitation affecting the sector, not information specific to Organon.

Payoff at expiry: unhedged vs. protective put vs. collar



Source: Black-Scholes priced on pre-leak realised volatility

Figure 3. Payoff at expiry - unhedged position, protective put, and collar.

A protective put struck near the unaffected price (\$6.90), priced using the correct pre-leak volatility regime (58.0%), costs \$0.08 per share. Pricing the same put using the post-announcement, deal-pinned volatility (4.5%) - a mistake this report deliberately tests for and rejects - would price it at effectively \$0.00, understating the true cost of protection by 100%. That collapse is not a rounding artefact: it is the direct, quantified consequence of using a volatility regime that only holds if the very event being insured against does not happen.

5. Recommendation

PASS on the unhedged spread at today's price - a conditional recommendation, not a categorical one.

The trade clears a reasonable risk-adjusted return only if the true probability of completion is close to the market's own implied level (93.9-94.9% across the break-price scenarios tested) rather than the more conservative ~92% mean assumption used in the Monte Carlo simulation. Two facts in this report lean toward the more optimistic reading: shareholder approval was already obtained on 23 July 2026, removing a major source of closing risk, and Sun Pharma's own stock reaction to the announcement was positive and statistically significant - not the reaction of a market that expects the deal to unravel or believes the acquirer overpaid.

Set against that: this report did not verify what regulatory or other closing conditions remain outstanding as of the snapshot date, beyond what is stated in the SEC filings reviewed. That is a genuine, unclosed gap in the analysis, not a rounding error, and it should be closed before any real capital is sized against either the market's implied probability or this report's own Monte Carlo assumption.

If a position is taken, the correctly-priced hedge available is a protective put struck near the unaffected price and priced at the pre-leak volatility regime (58.0%, \$0.08 per share) - not one calibrated to the deal-pinned volatility the market currently quotes, which would understate the true cost of protection by approximately 100%, as demonstrated in Section 4.4.

6. Limitations

- The DCF's discount rate is calibrated to reproduce the unaffected market price. This validates internal consistency - cross-checked against two independently modelled bounds, the embedded and marginal cost-of-debt estimates - but it is the market's own implied rate, not an independent, ground-up estimate of fair value arrived at without reference to the traded price.
- No dollar synergy figure was disclosed by Sun Pharma, confirmed directly against the primary SEC filing rather than assumed. Every synergy number in this report is this report's own reverse-DCF construction, not a management estimate.
- The break price used throughout (\$6.90, the unaffected close, with -10%/-20% sensitivity) is a proxy, not a measurement. A broken deal can print anywhere, including below the unaffected level if underlying fundamentals have deteriorated during the intervening process.
- Outstanding regulatory or other closing conditions as of the snapshot date were not verified beyond the SEC filings reviewed for this report.
- Organon's fiscal year (December) and Sun Pharma's (March) were reconciled to trailing-twelve-month figures where the two companies are compared directly; other same-period comparisons used each company's own most recent annual filing, which are not perfectly contemporaneous.
- This is a point-in-time analysis as of 6 August 2026. Prices, spreads, and implied probabilities move daily; the qualitative conclusions and methodology are intended to remain valid, but the specific figures reported will drift.

Bibliography

- Andrade, G., Mitchell, M., & Stafford, E. (2001). New evidence and perspectives on mergers. *Journal of Economic Perspectives*, 15(2), 103-120.
- Damodaran, A. (2026). Equity risk premiums (ERP): Determinants, estimates and implications - The 2026 edition. New York University Stern School of Business.
- Damodaran, A. Ratings, interest coverage ratios and default spreads [Data file]. New York University Stern School of Business. Retrieved August 2026, from https://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/smallrating.htm
- Jensen, M. C. (1986). Agency costs of free cash flow, corporate finance, and takeovers. *American Economic Review*, 76(2), 323-329.
- Mitchell, M., & Pulvino, T. (2001). Characteristics of risk and return in risk arbitrage. *Journal of Finance*, 56(6), 2135-2175.
- Organon & Co. (2026, April 27). Form 8-K, Exhibit 99.1: Sun Pharma signs definitive agreement to acquire Organon [Press release]. U.S. Securities and Exchange Commission, EDGAR, CIK 0001821825, accession no. 0001193125-26-178718.
- Roll, R. (1986). The hubris hypothesis of corporate takeovers. *Journal of Business*, 59(2), 197-216.
- Yahoo Finance. Historical price, volume, and financial statement data for OGN, SUNPHARMA.NS, ^GSPC, ^NSEI, ^VIX, ^TNX, INR=X, and peer tickers. Retrieved 6 August 2026.